

Course name: Biometric Security

Course Placement: Technical elective open to B.Tech. ICT Semester VI and VII, MSc (Data Science) and M.Tech sem II students

Course credits: LTP-C (3-0-2 -4)

Prerequisites:

- **Image Processing OR computer vision**
- **Machine Learning**
- **Python**

Course Outcome:

After the successful completion of the course the students should:

- develop an in-depth understanding of the biometric recognition systems and all associated aspects;
- be able to comprehend and present state-of-the-art research papers;
- be able to work on a specific research problem as part of the project work and,
- be able to write technical report

Course content:

Introduction to biometrics: basic processing pipeline, biometric recognition including identification and verification, biometric traits or modalities, specific characteristics of biometric traits, different biometric modes: face, speech, iris, palmprint, finger-vein, palm-vein, fingerprint. Evaluation metrics: impostor scores, genuine scores, FAR, FRR, EER, GAR, DET curve, CMC curve, Receiver operating characteristics, Face recognition: state-of-the-art in face biometrics, deep learning architecture pipeline, hand-crafted features, vascular biometrics: Acquisition, vascular biometric traits: palm-vein, finger-vein, dorsal hand-vein, iris. Attacks: Presentation attacks on biometric systems: real-world attack instances, different defense measures, Adversarial attacks using physical accessories, morphing attacks, multi-modal biometric systems: fusion at different levels: image-level fusion, score-level fusion, and decision-level fusion. Cancelable biometrics, secure and privacy preserving biometrics, face de-identification, cross-domain/heterogeneous recognition, Person re-identification.

Tentative schedule:

S.No.	Topic	No. of Lectures
1.	Introduction to Biometrics:	4
	Biometric traits: face, iris, fingerprint, palm, ear, gait, vascular traits, speech; biometric modes: verification and identification, biometric processing pipeline, evaluation metrics: impostor and genuine scores, equal error rate, DET curve, cumulative match characteristics curve, ROC curve	
2.	Relevant Image Processing basics	4
3.	Face recognition: overview and challenges	6
	Face verification and identification, challenges such as pose, illumination, occlusion; Handcrafted FR features, deep learning-based approaches	
4.	Fingerprint biometrics: overview and challenges	5

	Biometric acquisition, features: ridges, minutiae, pore; seminal fingerprint recognition techniques. Research on partial fingerprint matching	
5.	Vascular biometrics: overview and challenges:	4
	Acquisition, vascular biometric traits: palm-vein, finger-vein, dorsal hand-vein, iris	
6.	Multi-modal biometric systems:	2
	Fusion at different levels: image-level fusion, score-level fusion, decision-level fusion.	
7.	Research Presentations I	7
	Presentation attacks: real-world attack instances, different defense measures; adversarial attacks using digital/ physical accessories, poisoning attacks, morphing attacks	
8.	Research Presentations II	8
	cancelable biometrics: secure and privacy preserving biometrics, face de-identification cross-domain recognition: heterogenous person re-identification	

Grading policy:

Mid-sem exam: 15%

Paper presentation: 15%

Lab assignments: 20%

End-sem exam: 20%

Major project: presentation, report and viva: 30%

Textbook:

- i. A. Jain, A. Ross and K. Nandakumar, Introduction to Biometrics, first edition, Springer 2011, ISBN number: 978-0-387-77326-1

References:

- ii. P. Pathak, NPTEL course on Biometrics <https://nptel.ac.in/courses/106104119>
- iii. Representative research papers either for student presentations or as part of lecture material:
 1. A. Ross, S. Banerjee., C. Chen, A. Chowdhury, V. Mirjalili, R. Sharma, T. Swearingen. and S. Yadav, Some research problems in biometrics: The future beckons. In *IEEE International Conference on Biometrics (ICB)* (pp. 1-8), 2019.
 2. K. Raja, M. Ferrara, Franco, A., Spreuwers, L., Batskos, I., de Wit, F., Gomez-Barrero, M., Scherhag, U., Fischer, D., Venkatesh, S.K. and Singh, J.M., Morphing attack detection-database, evaluation platform, and benchmarking, *IEEE transactions on information forensics and security*, vol. 16, 2020, pp.4336-4351.
 3. U. Scherhag, C. Rathgeb, J. Merkle, and C. Busch, Deep face representations for differential morphing attack detection, *IEEE transactions on information forensics and security*, 15, 2020, pp.3625-3639.
 4. S. Venkatesh, R. Ramachandra, K. Raja and C. Busch, Face morphing attack generation and detection: A comprehensive survey. *IEEE transactions on technology and society*, 2(3), 2021, pp.128-145.

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